Guidance-Induced Leakage of Score Estimation Errors in Diffusion Models

Anonymous

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Abstract

Classifier-free guidance combines conditional and unconditional score estimates to steer diffusion sampling. The effect of score estimation error, however, depends on the probability law under which the error is evaluated. We study guidance-induced leakage: an error component that is weighted-solenoidal, and therefore invisible to marginal dynamics under its native law, can acquire an observable component under the guided law. A weighted Helmholtz–Hodge decomposition separates the inherited visible error from this newly observable contribution. We characterize leakage through an exact change-of-density identity and identify its intrinsic energy with a weighted negative Sobolev norm. We then describe its response to guidance through two channels: changes in the combination of native errors and changes in the guided reference measure. Finally, characteristic representations separate the native-visible and leakage contributions to the log-density correction induced by learned scores. Controlled Gaussian-mixture experiments illustrate how the reported leakage contribution can dominate at large guidance scales. Qualitative CIFAR-10 experiments document strong-guidance degradation and examine conservative score distillation as a possible intervention. These results identify a mechanism by which guidance changes the distributional relevance of score errors; they do not imply that leakage or sampling distortion must increase monotonically with guidance.

1 Introduction

Diffusion models have achieved remarkable success in image generation (Ho et al., 2020; Song & Ermon, 2019; Song et al., 2021). To further improve generation quality and conditional fidelity, guidance methods have been introduced (Dhariwal & Nichol, 2021), among which classifier-free guidance (CFG) has become a standard technique widely adopted in modern image generation systems (Ho & Salimans, 2022; Rombach et al., 2022; Saharia et al., 2022). Despite its effectiveness, increasing the guidance strength often comes at the cost of reduced sample diversity, oversaturation, and degraded image quality (Sadat et al., 2025; Pavasovic et al., 2026).

Classical analyses of score-based generative models relate sampling accuracy to the L^2 error of learned scores (Chen et al., 2023). Other studies have examined the geometric structure of learned score fields, including their conservativity and gauge freedom (Chao et al., 2023; Horvat & Pfister, 2024). More recently, (Khelifa et al., 2026) showed that certain score errors can leave marginal distributions unchanged, revealing that error magnitude alone does not determine sampling quality. However, these insights concern error observability relative to a given probability law. Under CFG, score estimates from different branches are combined under a new guided distribution. This raises questions:

How does guidance strength alter the observation of score errors, and whether this amplify some negative performance of CFG?

In this work, we identify a phenomenon termed *Leakage*, where score estimation errors that are invisible under their native distributions become observable under CFG. The key insight is that error observability is not an intrinsic property of the error field alone, but also depends on the underlying probability distribution. As CFG changes both the combination of conditional and unconditional score errors and the distribution under which they are evaluated, previously invisible components may acquire observable effects on the marginal dynamics. To characterize this phenomenon, we develop a geometric framework based on weighted Helmholtz–Hodge decompositions, separating the observable errors inherited from individual branches from those newly activated by guidance. This reveals that CFG does not merely amplify existing score estimation errors, but can fundamentally change which components of these errors influence the generative process.

We futher investigate the connection of leakage score error and the image degradation under strong guidance(Chung et al., 2025; Sadat et al., 2025; Pavasovic et al., 2026). Natively invisible errors may become additional sources of distortion in the guided sampling dynamics, potentially exacerbating the effects of strong guidance. We proposes the final distribution of CFG with learned score and seperate them with native or leakage score error, which demonstrates the source of degradation. A toy classifier free guidance model is also trained to demonstraite this phenomenon.

Contributions. We investigate how classifier-free guidance changes the geometric observability of score estimation errors and how these changes contribute to distributional distortion under guided sampling. Our main contributions are summarized as follows.

First, we identify *guidance-induced leakage*, a phenomenon in which score errors that are invisible under their native distributions become observable under CFG. We establish a weighted Helmholtz–Hodge framework that characterizes leakage at arbitrary fixed guidance strengths and quantifies its intrinsic energy.

Second, we characterize how leakage evolves with guidance strength and identify two underlying mechanisms: *branch mismatch* and *guided-law deformation*. We further show that, near the conditional baseline, the leakage field emerges at first order in the guidance displacement, while its intrinsic energy exhibits quadratic leading-order growth.

Third, we derive an analytical expression for the sampling distribution of CFG with learned scores, distinguishing the effects of native score errors and guidance-induced leakage. We further conduct controlled experiments with trained diffusion models to investigate how leakage changes with guidance strength and its potential contribution to the distributional distortion associated with strong guidance.

2 Related Work

Classifier-Free Guidance. Guidance methods improve conditional generation by steering diffusion sampling toward desired conditions. Classifier guidance uses gradients from an auxiliary classifier (Dhariwal & Nichol, 2021), while classifier-free guidance (CFG) combines conditional and unconditional score estimates without requiring a separate classifier (Ho & Salimans, 2022). CFG has become a standard component of modern image generation systems, including latent diffusion

models and large-scale text-to-image models (Rombach et al., 2022; Saharia et al., 2022). Our work focuses on the behavior of score estimation errors under this guided combination.

Theoretical Understanding of CFG. A growing body of work studies the distributional properties and sampling dynamics of CFG. Bradley and Nakkiran (Bradley & Nakkiran, 2025) show that CFG does not generally sample from the commonly assumed powered-product distribution and interpret its dynamics through a predictor–corrector perspective. Wu et al. (Wu et al., 2024) analyze diffusion guidance under Gaussian mixture models, establishing its effects on conditional confidence and sample diversity. More recently, Jiang and Ma (Jiang & Ma, 2026) derive analytic representations of CFG-induced distributions, while Pavasovic et al. (Pavasovic et al., 2026) investigate overshoot and shrinkage effects. These studies characterize the distributional behavior of guidance; our work instead investigates how guidance alters the geometric observability of score learning errors.

Strong Guidance Degradation and Mitigation. Large guidance scales can improve conditional alignment but often introduce reduced diversity, oversaturation, and generation artifacts. Dynamic thresholding (Saharia et al., 2022) mitigates saturation by rescaling predicted samples, while CFG++ (Chung et al., 2025) addresses off-manifold effects associated with conventional guidance. Adaptive Projected Guidance (APG) (Sadat et al., 2025) decomposes the guidance update into parallel and orthogonal components and suppresses the component primarily responsible for oversaturation. Autoguidance (Karras et al., 2024) instead uses a weaker version of the model as a reference to improve generation quality. These approaches modify guidance to improve sampling behavior. Our work provides a complementary perspective by studying additional distortion arising from previously invisible score estimation errors.

Score Estimation Errors and Geometric Observability. Classical convergence analyses of diffusion models relate sampling discrepancies to the L^2 error of learned scores (Chen et al., 2023). Other studies investigate the geometric properties of score fields, including conservativity (Chao et al., 2023) and gauge freedom (Horvat & Pfister, 2024). More recently, Khelifa et al. (Khelifa et al., 2026) establish that only the gradient component of a score error field affects marginal dynamics, while its weighted-solenoidal component remains invisible. Our work extends this geometric perspective to CFG, where conditional and unconditional score errors are combined under a different probability law. We characterize how such a change can activate natively invisible errors and introduce additional observable components into guided sampling.

3 Leakage at Fixed Guidance

The standard weighted-Hodge decomposition classifies score errors relative to the marginal law of a single diffusion process. However, classifier-free guidance involves three distinct probability laws: the conditional law p_t^c, the unconditional law p_t^u, and the law ρ_t^γ induced by the exact-score guided dynamics. An error that is invisible under its native distribution need not remain invisible after the two branches are combined and evaluated under ρ_t^γ. We characterize this transfer of observability at fixed guidance strengths.

3.1 CFG-Specific Hodge Decomposition

Under CFG, the combined score estimation error is

$$e_t^\gamma = \gamma e_t^c + (1-\gamma) e_t^u. \tag{3.1}$$

For each branch $b \in \{c, u\}$, let P_ρ denote the orthogonal projection in $L^2(\rho)$ onto the closed subspace of gradient fields. We decompose the native score error as

$$v_t^b = P_{p_t^b} e_t^b, \qquad w_t^b = (I - P_{p_t^b}) e_t^b. \tag{3.2}$$

Here, v_t^b is the native-visible error, while w_t^b is the native-solenoidal error satisfying $\nabla \cdot (p_t^b w_t^b) = 0$.

Their guided combinations are

$$\begin{aligned} v_t^\gamma &= \gamma v_t^c + (1-\gamma) v_t^u, \\ w_t^\gamma &= \gamma w_t^c + (1-\gamma) w_t^u, \end{aligned} \tag{3.3}$$

with $e_t^\gamma = v_t^\gamma + w_t^\gamma$.

We further project these components under the guided distribution:

$$\begin{aligned} h_t^\gamma &:= P_{\rho_t^\gamma} v_t^\gamma, \\ g_t^\gamma &:= P_{\rho_t^\gamma} w_t^\gamma, \\ o_t^\gamma &:= P_{\rho_t^\gamma} e_t^\gamma = h_t^\gamma + g_t^\gamma. \end{aligned} \tag{3.4}$$

We refer to h_t^γ as the inherited visible contribution and g_t^γ as *guidance-induced leakage*. This decomposition distinguishes errors already visible in their native branches from those made observable by guidance. It is not an orthogonal decomposition, since both components belong to the guided gradient subspace.

To quantify leakage, let $\dot{H}^1(\rho)$ be the weighted homogeneous Sobolev space equipped with $\|\psi\|_{\dot{H}^1(\rho)} = \|\nabla\psi\|_{L^2(\rho)}$, and let $\dot{H}^{-1}(\rho)$ denote its Hilbert dual. Define the weighted-divergence operator Q_ρ by

$$\langle Q_\rho z, \psi \rangle = \int_{\mathbb{R}^d} z \cdot \nabla\psi \, \rho \, dx. \tag{3.5}$$

Formally, $Q_\rho z = -\rho^{-1} \nabla \cdot (\rho z)$.

The leakage source and its intrinsic energy are defined as

$$q_t^\gamma := Q_{\rho_t^\gamma} w_t^\gamma, \qquad \mathcal{L}_t(\gamma) := \|g_t^\gamma\|^2_{L^2(\rho_t^\gamma)}. \tag{3.6}$$

3.2 Exact Characterization of Fixed-Guidance Leakage

Assumption 1 (Fixed-Guidance Regularity)**.** *For each fixed t and finite γ, the densities p_t^c, p_t^u, and ρ_t^γ are strictly positive, and the corresponding exact-score dynamics are well posed. The branch errors admit native weighted-Hodge decompositions, with all components reprojected under ρ_t^γ belonging to $L^2(\rho_t^\gamma)$. The log-density ratios $\log(\rho_t^\gamma / p_t^b)$ are locally C^1, and their directional derivatives along w_t^b are locally integrable under ρ_t^γ.*

Theorem 1 (Fixed-Guidance Leakage)**.** *Under Assumption 1, the guidance-induced leakage g_t^γ satisfies the following properties.*

(i) Exact observability-transfer identity. The leakage source is given by

$$q_t^\gamma = -\gamma w_t^c \cdot \nabla \log \frac{\rho_t^\gamma}{p_t^c} - (1-\gamma) w_t^u \cdot \nabla \log \frac{\rho_t^\gamma}{p_t^u}. \tag{3.7}$$

Moreover, $Q_{\rho_t^\gamma} g_t^\gamma = q_t^\gamma$.

(ii) Intrinsic leakage energy. The observable leakage energy satisfies

$$\mathcal{L}_t(\gamma) = \|g_t^\gamma\|^2_{L^2(\rho_t^\gamma)} = \|q_t^\gamma\|^2_{\dot{H}^{-1}(\rho_t^\gamma)}. \tag{3.8}$$

(iii) Zero-leakage criterion. Leakage vanishes if and only if

$$g_t^\gamma = 0 \iff q_t^\gamma = 0 \iff \nabla \cdot (\rho_t^\gamma w_t^\gamma) = 0. \tag{3.9}$$

In particular, $g_t^1 = 0$ *when* $\rho_t^1 = p_t^c$, *and* $g_t^0 = 0$ *when* $\rho_t^0 = p_t^u$.

Theorem 1 shows that errors invisible under their native distributions can become observable under CFG. In particular, Eq. (3.7) identifies the interaction between native-solenoidal errors and the density ratios ρ_t^γ / p_t^b as the source of leakage. Thus, leakage depends not only on the error fields themselves, but also on how guidance changes the reference probability law. It vanishes when the two weighted-divergence contributions cancel, including when each native-solenoidal field is tangent to the corresponding density-ratio level sets.

The intrinsic energy also admits a dual variational representation:

$$\mathcal{L}_t(\gamma) = \sup_{\psi \in \dot{H}^1(\rho_t^\gamma)} \left\{ 2\langle q_t^\gamma, \psi \rangle - \|\nabla \psi\|^2_{L^2(\rho_t^\gamma)} \right\}. \tag{3.10}$$

This characterization provides a variational objective for estimating leakage without explicitly solving the weighted Poisson equation. The corresponding Poisson and minimum-energy representations are given in the Appendix.

Leakage also has a direct interpretation in the guided marginal dynamics. Consider a perturbed probability-flow velocity $b_t^\gamma + \varepsilon \kappa_t e_t^\gamma$. Provided the first-order density response $\eta_t^\gamma = \partial_\varepsilon \rho_t^{\gamma,\varepsilon}|_{\varepsilon=0}$ exists, it satisfies

$$\partial_t \eta_t^\gamma + \nabla \cdot (b_t^\gamma \eta_t^\gamma) = -\kappa_t \nabla \cdot (\rho_t^\gamma e_t^\gamma). \tag{3.11}$$

The contribution from native-solenoidal errors is

$$-\kappa_t \nabla \cdot (\rho_t^\gamma w_t^\gamma) = \kappa_t \rho_t^\gamma q_t^\gamma. \tag{3.12}$$

Hence, guidance-induced leakage acts as an additional source of first-order marginal distortion, despite being invisible under the native distributions.

Having characterized leakage at a fixed guidance strength, we next investigate how it evolves as the guidance scale increases.

Corollary 1 (One-Way Observability Transfer)**.** *Under Assumption 1, suppose the native and guided densities are strictly positive and* C^1. *Then the native-visible error remains visible under the guided law:*

$$P_{\rho_t^\gamma} v_t^\gamma = v_t^\gamma, \qquad (I - P_{\rho_t^\gamma}) v_t^\gamma = 0. \tag{3.13}$$

Consequently, the total guided-visible error satisfies

$$o_t^\gamma = v_t^\gamma + g_t^\gamma. \tag{3.14}$$

Thus, native-invisible errors may become observable under CFG, whereas native-visible errors cannot become guided-invisible.

4 Leakage Evolution under Guidance

We now investigate how leakage evolves as the guidance strength varies. Recall that

$$g_t^\gamma = P_{\rho_t^\gamma} w_t^\gamma, \qquad w_t^\gamma = \gamma w_t^c + (1-\gamma) w_t^u. \tag{4.1}$$

There are two distinct sources of variation: changing γ modifies the combination of native-solenoidal errors, while simultaneously changing the guided distribution ρ_t^γ and its associated Hodge projection. We characterize these effects through a common-frame formulation and a weak two-channel response identity.

Common-Frame Formulation. As the guidance scale varies, the corresponding weighted Hodge projections act on different Hilbert spaces. To compare these projections, we introduce a common transport frame.

Fix a sampling time t, a compact guidance interval I, and a reference scale $\gamma_0 \in I$. Let Φ_t^γ denote the exact guided flow and define

$$\mu := \rho_t^{\gamma_0}, \qquad T_\gamma := \Phi_t^\gamma \circ (\Phi_t^{\gamma_0})^{-1}. \tag{4.2}$$

Then $(T_\gamma)_\# \mu = \rho_t^\gamma$.

Writing

$$J_\gamma := DT_\gamma, \quad B_\gamma := J_\gamma^{-1}, \quad A_\gamma := B_\gamma B_\gamma^\top, \tag{4.3}$$

we define the pulled-back leakage potential and error field by

$$u_\gamma := \phi_t^\gamma \circ T_\gamma, \qquad \beta_\gamma := B_\gamma (w_t^\gamma \circ T_\gamma), \tag{4.4}$$

where $g_t^\gamma = \nabla \phi_t^\gamma$.

The moving weighted-Poisson problems can then be expressed on the fixed reference space:

$$\int \nabla u_\gamma^\top A_\gamma \nabla v \, d\mu = \int \beta_\gamma \cdot \nabla v \, d\mu, \tag{4.5}$$

for every $v \in \dot{H}^1(\mu)$.

Assumption 2 (Uniform Transport-Frame Regularity)**.** *For every $\gamma \in I$, T_γ is a C^1 diffeomorphism, and there exist constants $0 < m \le M < \infty$ such that*

$$m|\xi| \le |J_\gamma(y)\xi| \le M|\xi| \tag{4.6}$$

uniformly in γ, for μ-almost every y. Moreover,

$$\begin{aligned} &\gamma \mapsto B_\gamma \in C^1(I; L^\infty(\mu)), \\ &\gamma \mapsto \beta_\gamma \in C^1(I; L^2(\mu)). \end{aligned} \tag{4.7}$$

Under Assumption 2, the solution u_γ of Eq. (4.5) is continuously differentiable in $\dot{H}^1(\mu)$. Consequently, the pulled-back leakage field

$$\widetilde{g}_t^\gamma := g_t^\gamma \circ T_\gamma = B_\gamma^\top \nabla u_\gamma \tag{4.8}$$

is continuously differentiable in $L^2(\mu)$. The detailed proof is provided in Appendix B.

The common frame establishes differentiability in a fixed Hilbert space, but does not automatically imply Eulerian differentiability at fixed spatial coordinates. To characterize the latter in a weak sense, we use the following moving-measure formulation.

Weak Guidance-Path Regularity. For a vector field F frozen at guidance scale γ, define

$$\mathfrak{m}_{t,\gamma}[F](\psi) := \frac{d}{dh}\Big|_{h=0} \int F \cdot \nabla\psi \, d\rho_t^{\gamma+h}. \tag{4.9}$$

Assumption 3 (Weak Guidance-Path Regularity)**.** *For each fixed t and γ in the interior of I, assume that the guided densities are strictly positive and continuously differentiable in γ in $L^2_{\mathrm{loc}}(\mathbb{R}^d)$.*

The native-solenoidal errors belong to $L^2(\rho_t^\gamma)$, and the guided leakage projections are locally continuous in the Eulerian coordinates:

$$g_t^{\gamma+h} \longrightarrow g_t^\gamma \quad \text{in } L^2_{\mathrm{loc}}(\mathbb{R}^d). \tag{4.10}$$

Finally, the moving-measure functional associated with the guided-solenoidal residual,

$$\mathcal{D}_{t,\gamma} := \mathfrak{m}_{t,\gamma}[w_t^\gamma - g_t^\gamma], \tag{4.11}$$

extends continuously to $\dot{H}^1(\rho_t^\gamma)$, namely, $\mathcal{D}_{t,\gamma} \in \dot{H}^{-1}(\rho_t^\gamma)$.

These conditions allow the weak parameter response to be obtained from finite differences of the Hodge orthogonality relations, without assuming a weak Leibniz rule. The corresponding argument and sufficient conditions for local projection continuity are given in Appendix B.

Two-Channel Leakage Response. We define the weak Eulerian response of g_t^γ as the gradient field $G_t^\gamma \in \mathcal{G}_{\rho_t^\gamma}$ characterized by

$$\langle G_t^\gamma, \nabla\psi\rangle_{\rho_t^\gamma} = \frac{d}{dh}\Big|_{h=0} \int g_t^{\gamma+h} \cdot \nabla\psi \, d\rho_t^{\gamma+h} - \mathfrak{m}_{t,\gamma}[g_t^\gamma](\psi). \tag{4.12}$$

The existence and representation of this response are established by the following theorem.

Theorem 2 (Two-Channel Leakage Response)**.** *Under Assumptions 2 and 3, the weak Eulerian response G_t^γ exists. For every $\psi \in \dot{H}^1(\rho_t^\gamma)$,*

$$\langle G_t^\gamma, \nabla\psi\rangle_{\rho_t^\gamma} = \langle \partial_\gamma w_t^\gamma, \nabla\psi\rangle_{\rho_t^\gamma} + \mathcal{D}_{t,\gamma}(\psi). \tag{4.13}$$

Let $D_t^\gamma \in \mathcal{G}_{\rho_t^\gamma}$ be the unique Riesz gradient representative of $\mathcal{D}_{t,\gamma}$, satisfying

$$\langle D_t^\gamma, \nabla\psi\rangle_{\rho_t^\gamma} = \mathcal{D}_{t,\gamma}(\psi), \tag{4.14}$$

and define

$$M_t^\gamma := P_{\rho_t^\gamma}(\partial_\gamma w_t^\gamma). \tag{4.15}$$

Then the leakage response admits the exact decomposition

$$G_t^\gamma = M_t^\gamma + D_t^\gamma. \tag{4.16}$$

For linear CFG,

$$M_t^\gamma = P_{\rho_t^\gamma}(w_t^c - w_t^u). \tag{4.17}$$

Moreover,

$$\|G_t^\gamma\|_{L^2(\rho_t^\gamma)} \le \|\partial_\gamma w_t^\gamma\|_{L^2(\rho_t^\gamma)} + \|\mathcal{D}_{t,\gamma}\|_{\dot{H}^{-1}(\rho_t^\gamma)}. \tag{4.18}$$

Main Theorem II identifies two mechanisms governing the evolution of guidance-induced leakage. The branch-combination channel M_t^γ captures the explicit change in the combination of conditional and unconditional native-solenoidal errors. In contrast, the moving-measure channel D_t^γ captures changes in error observability induced by the deformation of the guided reference law. The complete weak differentiation argument is provided in Appendix B.

5 Analytic Distribution of Learned CFG

We now connect guidance-induced leakage to the distribution generated by learned CFG. We first derive the distribution induced by learned scores and then separate the resulting correction into native-visible and leakage contributions.

Learned-score distribution. Recall that $\widehat{s}_t^b = s_t^b + e_t^b$ for $b \in \{c, u\}$, and

$$\widehat{b}_t^\gamma = a_t + \kappa_t(s_t^\gamma + e_t^\gamma), \tag{5.1}$$

where $s_t^\gamma = \gamma s_t^c + (1-\gamma)s_t^u$ and $e_t^\gamma = \gamma e_t^c + (1-\gamma)e_t^u$.

Let $H_t = \log(p_t^c/p_t^u)$, and let $\widehat{\Phi}_{T,t}^\gamma(x)$ denote the position at time t of the learned characteristic reaching x at time T.

Theorem 3 (Analytic Distribution of Learned CFG)**.** *Assume a common entrance law $p_0^c = p_0^u = \widehat{\rho}_0^\gamma$, well-posed learned characteristics, and the regularity required for the following path integrals, as specified in Appendix C. Then*

$$\widehat{\rho}_T^\gamma(x) = (p_T^c(x))^\gamma (p_T^u(x))^{1-\gamma} \exp\left[-\int_0^T \kappa_t \left\{ \gamma(\gamma-1)\|\nabla H_t\|^2 + \nabla \cdot e_t^\gamma + e_t^\gamma \cdot s_t^\gamma \right\} \left(\widehat{\Phi}_{T,t}^\gamma(x)\right) dt \right]. \tag{5.2}$$

The first term in the path-integral correction arises from ideal CFG, while the remaining terms are introduced by score estimation errors. All terms are evaluated along the learned sampling trajectory.

Native-visible and leakage corrections. To isolate the distributional effects of score estimation errors from those intrinsic to ideal guidance, we take the exact-score CFG distribution ρ_t^γ as the reference.

Recall that $e_t^\gamma = v_t^\gamma + w_t^\gamma$ and the leakage source from Main Theorem I is

$$q_t^\gamma = -\gamma w_t^c \cdot \nabla \log \frac{\rho_t^\gamma}{p_t^c} - (1-\gamma) w_t^u \cdot \nabla \log \frac{\rho_t^\gamma}{p_t^u}. \tag{5.3}$$

Theorem 4 (Exact Native–Leakage Distribution Decomposition)**.** *Under the assumptions of Theorem 3, suppose the ideal and learned CFG flows share the same initial distribution and the relevant sources are integrable along learned characteristics. Then*

$$\widehat{\rho}_T^\gamma(x) = \rho_T^\gamma(x) \exp\left[\int_0^T \kappa_t \left\{ -\nabla \cdot v_t^\gamma - v_t^\gamma \cdot \nabla \log \rho_t^\gamma + q_t^\gamma \right\} \left(\widehat{\Phi}_{T,t}^\gamma(x)\right) dt \right]. \tag{5.4}$$

The first two terms characterize the inherited native-visible error, while q_t^γ captures guidance-induced leakage. This is an exact decomposition of the log-density correction relative to ideal CFG, rather than an additive decomposition of KL divergence.

Two sources of leakage at fixed guidance. The leakage source can be further decomposed without differentiating with respect to the guidance strength:

$$q_t^\gamma = \gamma(1-\gamma)(w_t^c - w_t^u) \cdot \nabla H_t - w_t^\gamma \cdot \nabla \log \frac{\rho_t^\gamma}{(p_t^c)^\gamma (p_t^u)^{1-\gamma}}. \tag{5.5}$$

The first term represents *branch mismatch*, arising from the difference between conditional and unconditional native-solenoidal errors. The second represents *guided-law deviation*, capturing the discrepancy between the exact-score guided distribution and the instantaneous powered-product reference.

6 Simulation

In this section, we investigate guidance-induced leakage and its effects on synthetic distributions and pretrained diffusion models. We consider two synthetic distribution families, Gaussian mixture models (GMMs) and mixture of Gamma distributions, to examine the leakage mechanism in controlled settings. We also consider EDM to study its implications for image generation under classifier-free guidance (CFG). Add description for EDM.

6.1 Model Simulation

6.1.1 Gaussian Mixture Model

Experimental Setup. We construct a 20-dimensional Gaussian mixture model (GMM) with five equally weighted Gaussian components. Each component has a distinct mean and a full-rank covariance matrix. We select class 0 as the target distribution, while the unconditional distribution is defined as the mixture of all five components.

To investigate the observability of score estimation errors, we construct synthetic errors for both branches:

$$e_t^b(x) = \eta s_t^b(x) + \xi A_b s_t^b(x), \qquad b \in \{c, u\}, \tag{6.1}$$

where $s_t^b = \nabla \log p_t^b$ is the exact score, and η and ξ control the error magnitudes. To introduce controlled mismatch between branches, we define

$$\widetilde{A}_b = \sqrt{r}\, A_0 + \sqrt{1-r}\, B_b, \quad A_b = \sqrt{d}\, \frac{\widetilde{A}_b}{\|\widetilde{A}_b\|_F}, \qquad b \in \{c, u\} \tag{6.2}$$

where A_0, B_c, B_u are independently generated skew-symmetric matrices, each normalized to Frobenius norm $\sqrt{d}$. Add a claim that any vector field transformed via A_0, B_c, B_u would generate a non-gradient field. These matrices are constant in space and time. The parameter $r \in [0, 1]$ controls matrix sharing: $r = 1$ gives $A_c = A_u = A_0$, while smaller values introduce branch-specific components. We use $r = 1$ for the native branch baseline and $r = 0.95$ for the main CFG leakage experiment.

This construction provides an explicit decomposition into native-visible and native-invisible errors. Specifically, ηs_t^b is a gradient field, whereas $\xi A_b s_t^b$ satisfies $\nabla \cdot (p_t^b A_b s_t^b) = 0$ and is therefore invisible under the corresponding native distribution.

We evaluate the squared L^2 energies of both components along a 100-step noise schedule, reporting results every 10 steps from the high-noise endpoint to the data endpoint. The conditional energies are computed analytically, while the unconditional energies are estimated using Monte Carlo integration.

Further details on the distribution construction, error parameters, and numerical implementation are provided in the Appendix D.1.1.

Native Error Decomposition. We first evaluate the native visible and invisible score-error energies for the conditional and unconditional branches. For each branch $b \in \{c, u\}$, we define

$$\begin{aligned} E_{\text{vis}}^b(t) &= \mathbb{E}_{p_t^b}\left[\|\eta s_t^b\|^2\right], \\ E_{\text{inv}}^b(t) &= \mathbb{E}_{p_t^b}\left[\|\xi A_b s_t^b\|^2\right], \\ E_{\text{tot}}^b(t) &= E_{\text{vis}}^b(t) + E_{\text{inv}}^b(t). \end{aligned} \tag{6.3}$$

We report the invisible-energy ratio $R_{\text{inv}}^b(t) = E_{\text{inv}}^b(t)/E_{\text{tot}}^b(t)$.

Table 1: Native score-error energies along the sampling schedule with $r = 0.95$, averaged over five error realizations. The conditional branch corresponds to Class 0, while the unconditional branch uses the full five-component mixture. Step 0 is the high-noise endpoint and Step 100 is the generation endpoint.

		Conditional (Class 0)				Unconditional			
Step	λ	E_{vis}	E_{inv}	E_{tot}	R_{inv} (%)	E_{vis}	E_{inv}	E_{tot}	R_{inv} (%)
0	-20.000	0.020772	0.067667	0.088439	76.51	0.020777	0.067638	0.088415	76.50
10	-3.685	0.020747	0.067574	0.088320	76.51	0.020590	0.066993	0.087583	76.49
20	-2.248	0.020685	0.067341	0.088026	76.50	0.020203	0.065661	0.085864	76.47
30	-1.348	0.020623	0.067085	0.087708	76.49	0.019983	0.064911	0.084893	76.46
40	-0.639	0.020602	0.066947	0.087549	76.47	0.020152	0.065495	0.085647	76.47
50	0.000	0.020658	0.067035	0.087693	76.44	0.020702	0.067354	0.088056	76.49
60	0.639	0.020799	0.067389	0.088188	76.42	0.021490	0.069994	0.091484	76.51
70	1.348	0.021007	0.067955	0.088962	76.39	0.022360	0.072881	0.095241	76.52
80	2.248	0.021233	0.068587	0.089820	76.36	0.023159	0.075515	0.098674	76.53
90	3.685	0.021409	0.069088	0.090498	76.34	0.023730	0.077390	0.101120	76.53
100	20.000	0.021476	0.069280	0.090756	76.34	0.023939	0.078075	0.102014	76.53

Gaussian Mixture Model Results As shown in Table 1, the invisible component accounts for approximately 76.3–76.5% of the total native score-error energy in both branches across the reported noise levels. This proportion remains stable under the controlled branch mismatch ($r = 0.95$), providing a baseline for investigating how CFG changes the observability of native-invisible errors.

Table 2: Native (D_N) and leakage (D_L) KL contributions along the sampling schedule with $r = 0.95$. Here $D_N = \mathbb{E}[C_N]$ and $D_L = \mathbb{E}[C_L]$, whose sum gives $\mathrm{KL}(\widehat{\rho}_t^{\gamma} \| \rho_t^{\gamma})$. The variable λ denotes log-SNR. Entries are means over five error realizations, each evaluated using 512 Monte Carlo initial samples. The larger contribution within each pair is boldfaced. Guidance groups are defined for this experiment, with $\gamma = 1$ serving as the conditional baseline.

Step	λ	Baseline		Moderate guidance				Strong guidance			
		$\gamma = 1$		$\gamma = 3$		$\gamma = 5$		$\gamma = 10$		$\gamma = 20$	
		D_N	D_L	D_N	D_L	D_N	D_L	D_N	D_L	D_N	D_L
0	-20.000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
10	-3.685	**1.3871**	0.0000	**1.3898**	0.0004	**1.3926**	0.0059	**1.3989**	0.1336	1.3970	**2.1367**
20	-2.248	**1.6335**	0.0000	**1.6432**	0.0021	**1.6527**	0.0262	**1.6660**	0.5275	1.6506	**9.6647**
30	-1.348	**1.7820**	0.0000	**1.7989**	0.0052	**1.8103**	0.0593	**1.8158**	1.1093	1.9122	**32.0958**
40	-0.639	**1.8855**	0.0000	**1.9062**	0.0100	**1.9137**	0.1001	**1.9162**	1.7997	2.5507	**74.6623**
50	0.000	**1.9617**	0.0000	**1.9829**	0.0161	**1.9868**	0.1418	1.9935	**2.5313**	3.2721	**124.1010**
60	0.639	**2.0186**	0.0000	**2.0391**	0.0225	**2.0409**	0.1793	2.0552	**3.2401**	3.8351	**170.8518**
70	1.348	**2.0604**	0.0000	**2.0802**	0.0281	**2.0807**	0.2100	2.1032	**3.8645**	4.2103	**211.1399**
80	2.248	**2.0893**	0.0000	**2.1086**	0.0323	**2.1083**	0.2325	2.1378	**4.3510**	4.4207	**242.6010**
90	3.685	**2.1063**	0.0000	**2.1253**	0.0350	**2.1246**	0.2463	2.1588	**4.6594**	4.5064	**262.8207**
100	20.000	**2.1120**	0.0000	**2.1309**	0.0359	**2.1300**	0.2510	2.1658	**4.7650**	4.5283	**269.8599**

6.2 Experiments with Pretrained Diffusion Models

We investigate the practical implications of guidance-induced leakage in pretrained diffusion models, focusing on image generation under strong classifier-free guidance. To isolate the role of leakage, we remove it and observe how generation changes. Specifically, we construct a counterpart of the pretrained model in which the native-invisible errors of both branches are removed. Comparing the two models across guidance scales then shows what leakage does in a pretrained model.

Experimental Setup. We use publicly available class-conditional and unconditional EDM checkpoints (Karras et al., 2022) pretrained on CIFAR-10, which contains ten image classes at a resolution of 32×32. The two branches are independently trained networks. For each branch $b \in \{c, u\}$, the pretrained denoiser D^b defines the score

$$\widehat{s}_\sigma^b(x) = \frac{D^b(x, \sigma) - x}{\sigma^2}, \tag{6.4}$$

where σ is the noise level of the variance-exploding parameterization $x_\sigma = x_0 + \sigma\varepsilon$ and plays the role of the time index t in our analysis. The two branches are combined using

$$\widehat{s}_\sigma^\gamma = \gamma \widehat{s}_\sigma^c + (1 - \gamma)\widehat{s}_\sigma^u, \tag{6.5}$$

where $\gamma = 1$ corresponds to the conditional baseline. We refer to this original model as A.

Leakage-free distillation. For each branch, we fit the gradient of a scalar potential Φ_θ^b to the pretrained score under that branch's own noisy distribution:

$$\min_\theta \ \mathbb{E}_\sigma \, \mathbb{E}_{x \sim p_\sigma^b} \, \lambda(\sigma) \left\| \nabla_x \Phi_\theta^b(x, \sigma) - \widehat{s}_\sigma^b(x) \right\|^2. \tag{6.6}$$



Figure 1: CIFAR-10 samples under classifier-free guidance from the original EDM (left) and its leakage-free distilled counterpart (right), generated from the same initial noise and class labels (Heun, 40 steps). Each row is one class; two seeds per class.

Since $\nabla_x \Phi_\theta^b$ is a gradient field by construction, its best fit under p_σ^b is the native projection $P_{p_\sigma^b} \widehat{s}_\sigma^b = s_\sigma^b + v_\sigma^b$. The student keeps the ideal score and the native-visible error, and removes the native-invisible error w_σ^b. Because w_σ^b is invisible under the native distribution, this removal does not change generation at the conditional baseline. Moreover, the student remains a gradient field under any distribution, so $w^\gamma = 0$, and Theorem 1 guarantees $g^\gamma = 0$ at every guidance scale. The distilled model is thus free of guidance-induced leakage.

We first compare the original EDM with the variant in which both branches are replaced by their distilled counterparts. The detail of performance can be seen at D.2.2. We evaluate guidance scales $\gamma \in \{1, 3, 5, 7.5, 10, 12\}$ using deterministic Heun sampling with 40 steps ($\sigma_{\max} = 80$, $\sigma_{\min} = 0.002$, $\rho = 7$). All comparisons use matched initial noise, class labels, and image postprocessing, so that differences between samples reflect differences between the models.

Results. Figure 1 compares samples of the original and the distilled model generated from the same noise and labels. At $\gamma = 1$ the two models produce nearly identical samples, and up to $\gamma = 5$ they still generate the same objects with the same composition. Removing the native-invisible errors thus leaves generation essentially unchanged near the conditional baseline. As γ grows, the samples diverge in appearance, although most of them keep the same composition up to $\gamma = 7.5$. At $\gamma \geq 10$ both models break down, in qualitatively different ways: the original model saturates toward bleached, high-contrast images, whereas the distilled model develops high-frequency pixel artifacts and a color shift. Since the two models agree on everything the native training losses can see, this divergence shows that native-invisible errors become observable under strong guidance, as predicted by the leakage analysis.

7 Discussion

Guidance-induced leakage arises because weighted solenoidality depends on the reference density. A score error can have no marginal source under its native law and still contribute under the exact guided law. The fixed-guidance identity makes this dependence explicit, and the negative Sobolev norm measures the energy of the resulting observable projection. The guidance-response decomposition further distinguishes changes in branch combination from changes in the reference measure.

The density identities connect these geometric statements to learned sampling. They separate the exact-guidance correction from estimation error and, within the latter, separate native-visible and leakage contributions to a log-density ratio. This separation is useful for diagnosis, but it does not imply independent, nonnegative, or monotone contributions to a sampling discrepancy. Both terms depend on the learned characteristics, and their effects can interact.

The numerical evidence has a correspondingly limited scope. The controlled Gaussian-mixture example shows that the reported leakage contribution can become large at the strongest tested scales. The image experiments document degradation under strong guidance and show that conservative branch parameterizations can retain qualitative sample quality. They do not yet isolate leakage from changes in native-visible error or from discretization and model mismatch.

Several questions remain. Quantitative image experiments should measure both sample quality and changes in the visible error induced by distillation. Estimating the variational leakage energy in high-dimensional models would provide a more direct connection between the theory and the image observations. Relaxing the global transport-frame assumptions and establishing local guidance expansions would also clarify when the response formulas yield practical guidance-scale diagnostics.

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A Proofs for Fixed-Guidance Leakage

This appendix provides the functional-analytic details underlying Theorem 1 and Corollary 1. Section A.1 proves the fixed-guidance leakage identities and collects the variational and dynamical characterizations used in the main text. Section A.2 proves that native-visible errors cannot acquire a guided-solenoidal component under the stated regularity and cross-law integrability conditions.

A.1 Proof of Main Theorem I and Auxiliary Characterizations

Weighted Sobolev and Hodge spaces. Fix a strictly positive probability density ρ on $\mathbb{R}^d$. Let $\dot{H}^1(\rho)$ be the completion of $C_c^\infty(\mathbb{R}^d)$ under the gradient norm

$$\|\psi\|_{\dot{H}^1(\rho)} := \|\nabla\psi\|_{L^2(\rho)}, \tag{A.1}$$

where elements at zero gradient distance are identified. The gradient map extends by completion to an isometric isomorphism from $\dot{H}^1(\rho)$ onto the closed gradient subspace

$$\mathcal{G}_\rho := \overline{\{\nabla\psi : \psi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(\rho;\mathbb{R}^d)}. \tag{A.2}$$

Its orthogonal complement is $\mathcal{S}_\rho := \mathcal{G}_\rho^\perp$, which satisfies

$$\mathcal{S}_\rho = \{z \in L^2(\rho;\mathbb{R}^d) : \nabla\cdot(\rho z) = 0 \text{ in } \mathcal{D}'(\mathbb{R}^d)\}. \tag{A.3}$$

No global Poincaré inequality or $L^2(\rho)$ control of the potential is needed. We denote by P_ρ the orthogonal projection onto $\mathcal{G}_\rho$, and by $\dot{H}^{-1}(\rho)$ the Hilbert dual of $\dot{H}^1(\rho)$.

Define $Q_\rho : L^2(\rho;\mathbb{R}^d) \to \dot{H}^{-1}(\rho)$ and $L_\rho : \dot{H}^1(\rho) \to \dot{H}^{-1}(\rho)$ weakly by

$$\begin{aligned} \langle Q_\rho z, \psi\rangle &= \int z\cdot\nabla\psi\,\rho\,dx, \\ \langle L_\rho\phi, \psi\rangle &= \int \nabla\phi\cdot\nabla\psi\,\rho\,dx. \end{aligned} \tag{A.4}$$

The operator L_ρ is the Riesz isomorphism associated with $\dot{H}^1(\rho)$.

Lemma 1 (Weighted Poisson representation)**.** *For every $z \in L^2(\rho;\mathbb{R}^d)$, there exists a unique $\phi_z \in \dot{H}^1(\rho)$ satisfying*

$$\int \nabla\phi_z\cdot\nabla\psi\,\rho\,dx = \int z\cdot\nabla\psi\,\rho\,dx \quad \forall\psi \in \dot{H}^1(\rho). \tag{A.5}$$

Moreover,

$$P_\rho z = \nabla\phi_z, \qquad L_\rho\phi_z = Q_\rho z. \tag{A.6}$$

Proof. The functional $\ell_z(\psi) = \int z\cdot\nabla\psi\,\rho\,dx$ is bounded on $\dot{H}^1(\rho)$ by Cauchy–Schwarz:

$$|\ell_z(\psi)| \le \|z\|_{L^2(\rho)}\|\psi\|_{\dot{H}^1(\rho)}.$$

The Riesz representation theorem gives a unique ϕ_z satisfying (A.5). It follows that

$$\int (z - \nabla\phi_z)\cdot\nabla\psi\,\rho\,dx = 0 \quad \forall\psi \in \dot{H}^1(\rho).$$

Thus $z - \nabla\phi_z \in \mathcal{S}_\rho$ and $\nabla\phi_z \in \mathcal{G}_\rho$, proving $P_\rho z = \nabla\phi_z$. The operator definitions yield the weighted Poisson identity. □

Change of reference measure. The source of leakage is obtained by transporting a native-solenoidal field from its original weighted space to the guided one.

Lemma 2 (Weighted divergence under a change of density)**.** *Let p and ρ be strictly positive densities and put $r = \rho/p$. Suppose $r \in C^1_{\mathrm{loc}}(\mathbb{R}^d)$, $w \in L^1_{\mathrm{loc}}(\rho;\mathbb{R}^d)$, $w \cdot \nabla \log r \in L^1_{\mathrm{loc}}(\rho)$, and $\nabla \cdot (pw) = 0$ in $\mathcal{D}'(\mathbb{R}^d)$. Then*

$$-\nabla \cdot (\rho w) = -\rho w \cdot \nabla \log \frac{\rho}{p} \quad \text{in } \mathcal{D}'(\mathbb{R}^d). \tag{A.7}$$

Equivalently, $Q_\rho w = -w \cdot \nabla \log(\rho/p)$ in the weighted weak sense.

Proof. For $\psi \in C^\infty_c(\mathbb{R}^d)$, native solenoidality allows $r\psi$ as a test function, by local approximation if needed. Expanding its gradient gives

$$\begin{aligned} 0 &= \int pw \cdot \nabla(r\psi)\, dx \\ &= \int \rho w \cdot \nabla\psi \, dx + \int \rho\psi \, w \cdot \nabla \log r \, dx. \end{aligned} \tag{A.8}$$

Rearranging proves the distributional identity. All terms are locally integrable by the stated assumptions. □

Proof of fixed-guidance leakage.

Proof of Theorem 1. Fix t and γ, and abbreviate $\rho = \rho_t^\gamma$, $w = w_t^\gamma$, $g = g_t^\gamma$, and $q = q_t^\gamma$. By definition, $g = P_\rho w$, so $w - g \in \mathcal{S}_\rho$. Consequently,

$$Q_\rho w = Q_\rho g = q. \tag{A.9}$$

Lemma 1 gives a unique $\phi_t^\gamma \in \dot{H}^1(\rho_t^\gamma)$ with

$$g_t^\gamma = \nabla\phi_t^\gamma, \qquad L_{\rho_t^\gamma}\phi_t^\gamma = q_t^\gamma. \tag{A.10}$$

For each branch $b \in \{c, u\}$, native solenoidality gives $\nabla \cdot (p_t^b w_t^b) = 0$. Applying Lemma 2 with $p = p_t^b$ and $\rho = \rho_t^\gamma$ yields

$$Q_{\rho_t^\gamma} w_t^b = -w_t^b \cdot \nabla \log \frac{\rho_t^\gamma}{p_t^b}.$$

By linearity of $Q_{\rho_t^\gamma}$ and $w_t^\gamma = \gamma w_t^c + (1-\gamma) w_t^u$, we obtain the exact identity in (3.7).

Next, the weak Poisson identity implies

$$\langle q, \psi \rangle = \int \nabla\phi_t^\gamma \cdot \nabla\psi \, \rho \, dx \quad \forall \psi \in \dot{H}^1(\rho). \tag{A.11}$$

Taking the dual norm and applying Cauchy–Schwarz gives $\|q\|_{\dot{H}^{-1}(\rho)} \le \|\nabla\phi_t^\gamma\|_{L^2(\rho)}$. Equality follows by choosing $\psi = \phi_t^\gamma / \|\nabla\phi_t^\gamma\|_{L^2(\rho)}$ when the denominator is nonzero; the zero case is immediate. Thus

$$\|q\|^2_{\dot{H}^{-1}(\rho)} = \|g\|^2_{L^2(\rho)} = \mathcal{L}_t(\gamma), \tag{A.12}$$

which proves (3.8).

It follows that $g = 0$ if and only if $q = 0$. The weak definition of Q_ρ and (A.3) further give

$$q = 0 \iff \nabla \cdot (\rho w) = 0 \iff w \in \mathcal{S}_\rho,$$

proving (3.9). At $\gamma = 1$, if the guided and conditional laws share the initialization and the corresponding continuity equations have unique solutions, $\rho_t^1 = p_t^c$ and $w_t^1 = w_t^c \in \mathcal{S}_{p_t^c}$; therefore $q_t^1 = g_t^1 = 0$. Similarly, if $\rho_t^0 = p_t^u$, then $q_t^0 = g_t^0 = 0$. □

Equivalent energy representations. For completeness, the intrinsic energy admits both a dual variational formula and a minimum-energy characterization:

$$\begin{aligned} \mathcal{L}_t(\gamma) &= \langle q_t^\gamma, L_{\rho_t^\gamma}^{-1} q_t^\gamma \rangle \\ &= \sup_{\psi \in \dot{H}^1(\rho_t^\gamma)} \left\{ 2\langle q_t^\gamma, \psi \rangle - \|\nabla \psi\|_{L^2(\rho_t^\gamma)}^2 \right\}, \end{aligned} \tag{A.13}$$

$$\mathcal{L}_t(\gamma) = \inf_{\substack{z \in L^2(\rho_t^\gamma; \mathbb{R}^d) \\ Q_{\rho_t^\gamma} z = q_t^\gamma}} \|z\|_{L^2(\rho_t^\gamma)}^2. \tag{A.14}$$

To prove these identities, write $\phi = L_\rho^{-1} q$ and $g = \nabla \phi$. The Poisson equation gives $\langle q, \phi \rangle = \|g\|_{L^2(\rho)}^2$. For any $\psi \in \dot{H}^1(\rho)$,

$$2\langle q, \psi \rangle - \|\nabla \psi\|_{L^2(\rho)}^2 = \|g\|_{L^2(\rho)}^2 - \|\nabla \psi - g\|_{L^2(\rho)}^2. \tag{A.15}$$

Hence the supremum is attained at $\psi = \phi$. For the infimum, any feasible z satisfies $Q_\rho(z - g) = 0$, so $z - g \in \mathcal{S}_\rho$. By orthogonality,

$$\|z\|_{L^2(\rho)}^2 = \|g\|_{L^2(\rho)}^2 + \|z - g\|_{L^2(\rho)}^2.$$

The unique minimizer is therefore $z = g$. These identities justify the variational critic objective in (3.10).

Leakage in the linearized marginal dynamics. Consider the perturbed continuity equation

$$\partial_t \rho_t^{\gamma,\varepsilon} + \nabla \cdot [\rho_t^{\gamma,\varepsilon} (b_t^\gamma + \varepsilon \kappa_t e_t^\gamma)] = 0, \quad \rho_0^{\gamma,\varepsilon} = \pi_0. \tag{A.16}$$

Assume the derivative $\eta_t^\gamma = \partial_\varepsilon \rho_t^{\gamma,\varepsilon}|_{\varepsilon=0}$ exists and the weak continuity equation can be differentiated. Testing with a compactly supported smooth function and differentiating at $\varepsilon = 0$ yields

$$\partial_t \eta_t^\gamma + \nabla \cdot (b_t^\gamma \eta_t^\gamma) = -\kappa_t \nabla \cdot (\rho_t^\gamma e_t^\gamma), \qquad \eta_0^\gamma = 0. \tag{A.17}$$

Because $e_t^\gamma = v_t^\gamma + w_t^\gamma$ and $Q_{\rho_t^\gamma} w_t^\gamma = Q_{\rho_t^\gamma} g_t^\gamma$, the native-solenoidal contribution to this first-order source is

$$-\kappa_t \nabla \cdot (\rho_t^\gamma w_t^\gamma) = -\kappa_t \nabla \cdot (\rho_t^\gamma g_t^\gamma). \tag{A.18}$$

When q_t^γ has a function representation, this is $\kappa_t \rho_t^\gamma q_t^\gamma$. This establishes the dynamical interpretation used in (3.11)–(3.12).

A.2 Proof of One-Way Observability Transfer

We now establish the cross-law stability result underlying Corollary 1. The central point is that a gradient field in the closed weighted gradient space for one positive density remains a gradient field for another positive density, provided its energy under the second density is finite.

Lemma 3 (Cross-law stability of gradient fields)**.** *Let p and ρ be strictly positive C^1 probability densities on $\mathbb{R}^d$. If*

$$v \in \mathcal{G}_p, \qquad v \in L^2(\rho; \mathbb{R}^d), \tag{A.19}$$

then $v \in \mathcal{G}_\rho$.

Proof. Since $v \in \mathcal{G}_p$, there exists $\{\phi_n\} \subset C_c^\infty(\mathbb{R}^d)$ such that

$$\nabla\phi_n \longrightarrow v \quad \text{in } L^2(p; \mathbb{R}^d). \tag{A.20}$$

We establish the claim in four steps.

Step 1: Recover a distributional potential. For any compact $K \subset \mathbb{R}^d$, the strict positivity and continuity of p give $c_K := \inf_K p > 0$. Thus

$$\int_K |\nabla\phi_n - v|^2 \, dx \le c_K^{-1} \|\nabla\phi_n - v\|^2_{L^2(p)} \to 0.$$

Therefore $\nabla\phi_n \to v$ in $L^2_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^d)$. Since each $\nabla\phi_n$ is curl-free, the distributional limit satisfies

$$\partial_i v_j - \partial_j v_i = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^d).$$

The weak Poincaré lemma on the simply connected domain $\mathbb{R}^d$ yields $\Phi \in W^{1,2}_{\mathrm{loc}}(\mathbb{R}^d)$ with

$$v = \nabla\Phi \quad \text{a.e. on } \mathbb{R}^d. \tag{A.21}$$

Step 2: Truncate the potential. Let T_M denote truncation to $[-M, M]$, and set $\Phi_M = T_M(\Phi)$. By the Sobolev chain rule, $\nabla\Phi_M = \mathbf{1}_{\{|\Phi|<M\}} v$ almost everywhere. Since $v \in L^2(\rho; \mathbb{R}^d)$,

$$\|\nabla\Phi_M - v\|^2_{L^2(\rho)} = \int_{\{|\Phi|\ge M\}} |v|^2 \rho \, dx \longrightarrow 0. \tag{A.22}$$

Step 3: Introduce a compact cutoff. Choose $\chi_R \in C_c^\infty(\mathbb{R}^d)$ such that $0 \le \chi_R \le 1$, $\chi_R = 1$ on B_R, $\chi_R = 0$ outside B_{2R}, and $|\nabla\chi_R| \le C/R$. Set $\Phi_{M,R} = \chi_R \Phi_M$. Then

$$\nabla\Phi_{M,R} = \chi_R \nabla\Phi_M + \Phi_M \nabla\chi_R.$$

For fixed M, dominated convergence gives $\chi_R \nabla\Phi_M \to \nabla\Phi_M$ in $L^2(\rho)$. The cutoff term satisfies

$$\|\Phi_M \nabla\chi_R\|^2_{L^2(\rho)} \le \frac{C^2 M^2}{R^2} \int_{B_{2R}\setminus B_R} \rho \, dx \le \frac{C^2 M^2}{R^2} \longrightarrow 0. \tag{A.23}$$

Therefore $\nabla\Phi_{M,R} \to \nabla\Phi_M$ in $L^2(\rho)$ as $R \to \infty$ for each fixed M.

Step 4: Smooth and pass to the limit. For fixed M, R, the compactly supported function $\Phi_{M,R}$ belongs to $W^{1,2}(\mathbb{R}^d)$. Standard mollification produces $\Phi_{M,R,\delta} \in C_c^\infty(\mathbb{R}^d)$ such that $\nabla\Phi_{M,R,\delta} \to \nabla\Phi_{M,R}$ in unweighted L^2 as $\delta \to 0$. All functions are supported in a fixed compact set for sufficiently

small δ, on which ρ is bounded above. Consequently, the convergence also holds in $L^2(\rho)$. Combining this approximation with (A.22) and (A.23), and taking a diagonal sequence, yields $\psi_n \in C_c^\infty(\mathbb{R}^d)$ with

$$\nabla\psi_n \longrightarrow v \quad \text{in } L^2(\rho;\mathbb{R}^d).$$

Hence $v \in \mathcal{G}_\rho$ by definition. □

Proof of Corollary 1. For each $b \in \{c, u\}$, the native decomposition gives $v_t^b = P_{p_t^b} e_t^b \in \mathcal{G}_{p_t^b}$. The cross-law integrability assumption ensures $v_t^b \in L^2(\rho_t^\gamma;\mathbb{R}^d)$. Applying Lemma 3 with $p = p_t^b$ and $\rho = \rho_t^\gamma$ yields

$$v_t^c, v_t^u \in \mathcal{G}_{\rho_t^\gamma}.$$

Since $\mathcal{G}_{\rho_t^\gamma}$ is a linear subspace,

$$v_t^\gamma = \gamma v_t^c + (1-\gamma) v_t^u \in \mathcal{G}_{\rho_t^\gamma}.$$

It follows that

$$P_{\rho_t^\gamma} v_t^\gamma = v_t^\gamma, \qquad (I - P_{\rho_t^\gamma}) v_t^\gamma = 0. \tag{A.24}$$

Finally, by linearity of the guided projection,

$$o_t^\gamma = P_{\rho_t^\gamma}(v_t^\gamma + w_t^\gamma) = v_t^\gamma + g_t^\gamma.$$

This proves Corollary 1. □

The result concerns the direction of transfer between weighted gradient and solenoidal subspaces. It does not imply that the norm of the total guided-visible error is monotone in the guidance strength: v_t^γ and g_t^γ may still partially cancel.

B Proof of two channel theorem

C Detailed Derivation of the Learned-CFG Distribution

We provide full proofs of Theorems 3, following the characteristic-based derivation of ideal CFG in (Jiang & Ma, 2026).

D Simulation Details

D.1 Settings Detail

D.1.1 Simulation settings

Diffusion Settings. For both the GMM and Gamma-family experiments, we adopt the variance-preserving Gaussian perturbation process with the cosine-inspired noise schedule used in classifier-free guidance (Ho & Salimans, 2022). Given a clean sample $X_{\text{data}} \sim p_{\text{data}}$, the noise-perturbed sample is defined as

$$X_t = \alpha_t X_{\text{data}} + \sigma_t \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I_d), \tag{D.1}$$

where $\alpha_t^2 + \sigma_t^2 = 1$.

We parameterize the noise schedule using

$$\alpha_t = \cos\theta_t, \quad \sigma_t = \sin\theta_t, \quad \theta_t = (1-t)\arctan(e^{10}) + t\arctan(e^{-10}). \tag{D.2}$$

Here, $t \in [0,1]$ denotes sampling progress from the high-noise endpoint to the data endpoint.

The corresponding log-signal-to-noise ratio (log-SNR) is

$$\lambda_t = \log\frac{\alpha_t^2}{\sigma_t^2}, \tag{D.3}$$

which increases from -20 to 20 along the sampling direction.

We discretize the schedule into 100 intervals, yielding 101 evaluation points $t_n = n/100$, $n = 0, \ldots, 100$. For the native score-error analysis, we report the results every 10 steps.

The finite log-SNR endpoints provide close approximations to the standard Gaussian prior and the original data distribution, respectively. All noise-perturbed distributions are evaluated using their exact finite-noise definitions.

Gaussian Mixture Model Generation. We construct a 20-dimensional Gaussian mixture model (GMM) consisting of five Gaussian components with equal mixture weights. The unconditional data distribution is defined as

$$p^u_{\text{data}}(x) = \frac{1}{5}\sum_{c=0}^{4}\mathcal{N}(x;\mu_c,\Sigma_c), \tag{D.4}$$

where each component represents a conditional distribution

$$p^c_{\text{data}}(x) = \mathcal{N}(x;\mu_c,\Sigma_c). \tag{D.5}$$

We select Class 0 as the target conditional distribution throughout the experiments.

Component Means. We set the mean of Class 0 to the origin, $\mu_0 = \mathbf{0}$. For the remaining four components, we independently sample $z_c \sim \mathcal{N}(0, I_{20})$ and define

$$\mu_c = 4\frac{z_c}{\|z_c\|_2}, \qquad c = 1, \ldots, 4. \tag{D.6}$$

Thus, the remaining component means are randomly distributed on a sphere of radius 4 centered at the origin.

Component Covariances. Each component is assigned a full-rank covariance matrix

$$\Sigma_c = Q_c\,\mathrm{diag}(\nu_{c,1}, \ldots, \nu_{c,20})Q_c^\top, \tag{D.7}$$

where Q_c is a random orthogonal matrix obtained from the QR decomposition of a 20×20 matrix with independent standard Gaussian entries, with column signs adjusted to make the diagonal of the QR triangular factor nonnegative. The eigenvalues are independently sampled as

$$\nu_{c,j} \sim \mathcal{U}(0.5, 1.5), \qquad j = 1, \ldots, 20. \tag{D.8}$$

This construction ensures that all covariance matrices are symmetric positive definite, while allowing different orientations and variances across components. The geometry seed is fixed at 100.

Gamma Exponential Family Generation. For the second synthetic distribution family, we construct a 20-dimensional mixture of five product Gamma components, matching the GMM in dimension, component count, and uniform mixture weights. Let

$$g(z;k,\vartheta)=\frac{z^{k-1}e^{-z/\vartheta}}{\Gamma(k)\vartheta^k}\,\mathbf{1}_{\{z>0\}},\qquad k,\vartheta>0, \tag{D.9}$$

denote the Gamma density with shape k and scale ϑ. The conditional and unconditional data distributions are

$$\begin{aligned} p^c_{\text{data}}(x)&=\prod_{j=1}^{20} g(x_j;k_{c,j},\vartheta_{c,j}),\\ p^u_{\text{data}}(x)&=\frac{1}{5}\sum_{c=0}^{4} p^c_{\text{data}}(x). \end{aligned} \tag{D.10}$$

Different components have different shape and scale parameters. As in the GMM experiment, Class 0 is the target conditional component, while the unconditional branch is the equal-weight mixture. The component parameters remain fixed across all diffusion timesteps.

Exact Scores. **GMM** Under Gaussian perturbation, each GMM conditional distribution remains Gaussian:

$$p^c_t(x)=\mathcal{N}(x;\alpha_t\mu_c,C_{c,t}),\qquad C_{c,t}=\alpha_t^2\Sigma_c+\sigma_t^2 I_d. \tag{D.11}$$

The corresponding conditional score is

$$s^c_t(x)=-C^{-1}_{c,t}(x-\alpha_t\mu_c). \tag{D.12}$$

Gamma family. The Gaussian perturbation of a Gamma component is generally not Gamma distributed. Since the coordinates are independent within each component, its noised conditional density can nevertheless be written as a product of one-dimensional convolutions:

$$\begin{aligned} p^c_t(x)&=\prod_{j=1}^{20} h_{c,j,t}(x_j),\\ h_{c,j,t}(y)&=\int_0^\infty g(z;k_{c,j},\vartheta_{c,j})\varphi_{\sigma_t}(y-\alpha_t z)\,dz, \end{aligned} \tag{D.13}$$

where $\varphi_\sigma(y)$ is the density of $\mathcal{N}(0,\sigma^2)$. The conditional score is given coordinatewise by

$$s^c_{t,j}(x)=\frac{\displaystyle\int_0^\infty \frac{\alpha_t z-x_j}{\sigma_t^2}g(z;k_{c,j},\vartheta_{c,j})\varphi_{\sigma_t}(x_j-\alpha_t z)\,dz}{h_{c,j,t}(x_j)}. \tag{D.14}$$

Thus, its exact score admits a one-dimensional integral representation, which can be evaluated numerically without training a score network.

Unconditional mixtures. For either distribution family, the unconditional score is the mixture of its conditional scores weighted by the posterior component probabilities:

$$s^u_t(x)=\sum_{c=0}^{4}\frac{p^c_t(x)}{\sum_{j=0}^4 p^j_t(x)}s^c_t(x). \tag{D.15}$$

Synthetic Score Error Construction. For both the GMM and Gamma family, and for each branch $b \in \{c, u\}$, we construct the score estimation error as

$$e_t^b(x) = \eta s_t^b(x) + \xi A s_t^b(x), \tag{D.16}$$

where $\eta \sim \mathcal{U}(-0.05, 0.05)$ and $\xi \sim \mathcal{U}(0.04, 0.08)$. The skew-symmetric matrix is generated by

$$A = \sqrt{d}\,\frac{G - G^\top}{\|G - G^\top\|_F}, \qquad G_{ij} \sim \mathcal{N}(0, 1). \tag{D.17}$$

The parameters η, ξ, and A are shared across both branches and all timesteps within each random seed.

The term ηs_t^b is a gradient field, while $\xi A s_t^b$ is weighted-solenoidal under the native distribution, satisfying $\nabla \cdot (p_t^b A s_t^b) = 0$. Therefore, the two terms provide an exact native visible–invisible decomposition in both distribution families.

Numerical Evaluation. For both distribution families, we evaluate the squared L^2 energies of the native visible and invisible components every 10 steps along the 100-step schedule.

GMM. For conditional Gaussian distributions, the energies are computed analytically as

$$\begin{aligned} E_{\text{vis}}^c(t) &= \eta^2 \operatorname{tr}(C_{c,t}^{-1}), \\ E_{\text{inv}}^c(t) &= \xi^2 \operatorname{tr}(A C_{c,t}^{-1} A^\top). \end{aligned} \tag{D.18}$$

For the unconditional GMM, we estimate corresponding expectations using 65,536 independent Monte Carlo samples at each evaluation timestep.

Gamma family. The corresponding conditional and unconditional energies are defined by the same L^2 expectations, but do not have the GMM trace formulas. Their numerical evaluation uses the one-dimensional Gamma–Gaussian convolution scores defined above and Monte Carlo expectations over the respective noised distributions.

D.2 Performance on EDM

D.2.1 Strong Guidance Degradation Under EDM

We first reproduce strong guidance degradation using pretrained EDM models on CIFAR-10. Figure 2 presents automobile samples generated with DDIM and Heun across guidance strengths $\gamma \in \{1, 3, 5, 10, 20\}$. For each seed, the class label and initial noise are shared across guidance strengths and samplers, allowing a direct qualitative comparison.

Moderate guidance can sharpen vehicle features and improve visual clarity relative to the conditional baseline at $\gamma = 1$. However, increasing guidance to $\gamma = 10$ introduces pronounced color saturation and loss of fine detail. At $\gamma = 20$, many samples are dominated by saturated color regions, and the vehicle structure becomes difficult to recognize. This qualitative trend appears under both DDIM and Heun, demonstrating strong guidance degradation across the two sampling methods in this setting.

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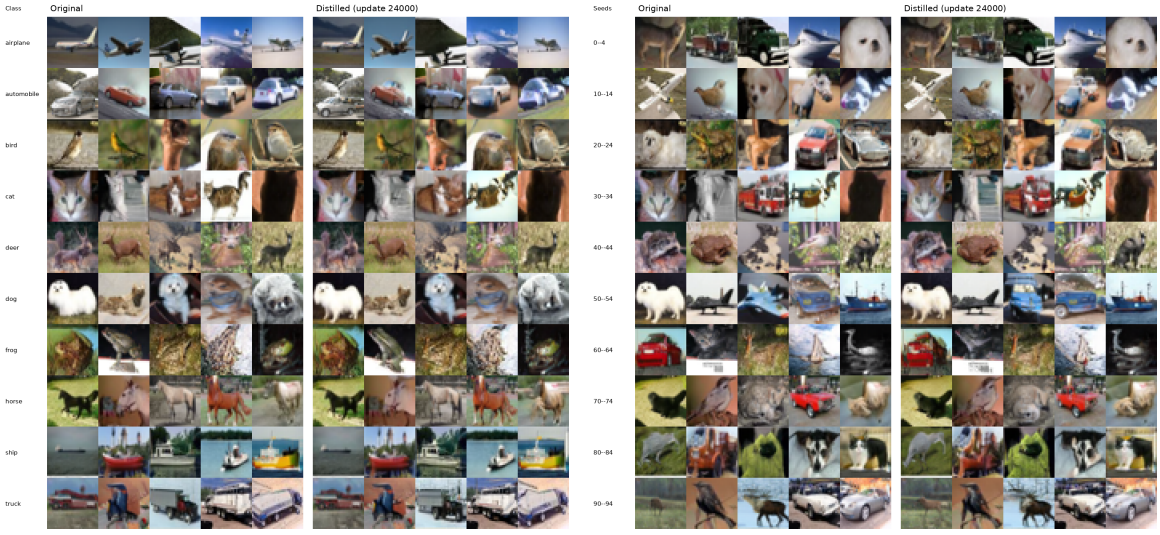
To further examine this behavior across classes, Figure 3 presents samples from all ten CIFAR-10 classes under DDIM sampling. The same initial noise and class labels are used across guidance

strengths. Strong guidance produces pronounced color saturation and loss of recognizable structure across multiple classes, extending the observations from the automobile examples.

D.2.2 Generation Quality of Distilled Branches

Before combining the distilled branches under CFG, we compare their generation quality with that of the original pretrained models. Figure 4 presents matched samples from the conditional and unconditional branches. All samples use deterministic Heun sampling with 40 steps. Within each branch, the original and distilled models share the same initial noise and, where applicable, the same class labels. The conditional branch is evaluated at the $\gamma = 1$ baseline, while the unconditional branch is sampled without class labels.

By construction, each distilled score is the gradient of a scalar potential. Since the ideal score is also a gradient field, the resulting score error has no native-invisible component under the assumptions of our weighted-Hodge framework. The displayed samples show that both distilled branches largely preserve the visual quality of their original counterparts, with some local differences in shape and texture. This provides qualitative evidence that conservative parameterization can retain branch-level generation quality, although it does not establish that the native-visible error remains unchanged.



(a) Conditional branch

(b) Unconditional branch

Figure 4: Generation quality before and after conservative score distillation using 40-step Heun sampling. Each panel compares 50 original samples (left) with 50 distilled samples (right). The conditional panel contains five matched sample pairs per CIFAR-10 class, with one class per row. Rows in the unconditional panel group matched seeds without prescribed class assignments. Initial Gaussian noise and applicable class labels are matched between the original and distilled models.

D.2.3 Performance on High guided field

. We construct the distilled model from the basemodel EDM to diminish the invisible part in condition and uncondition branch.



(a) DDIM (b) Heun

Figure 2: Strong guidance degradation of pretrained EDM on CIFAR-10 under DDIM and Heun, each using 40 sampling steps. All images use the automobile condition. Columns correspond to $\gamma = 1, 3, 5, 10, 20$, and rows correspond to seeds 10–19. The same initial noise is used for each seed across guidance strengths and samplers. Strong guidance produces severe color saturation and loss of visual detail in both panels.



(a) $\gamma = 1$ (b) $\gamma = 5$

(c) $\gamma = 10$ (d) $\gamma = 20$

Figure 3: Generated samples across all ten CIFAR-10 classes using pretrained EDM with 40-step DDIM sampling. Each panel contains 100 samples, arranged with one class per row and ten samples per class. Sample positions, initial noise, and class labels are matched across panels. Increasing guidance to $\gamma = 10$ and 20 leads to substantial color saturation and loss of visual detail across multiple classes.